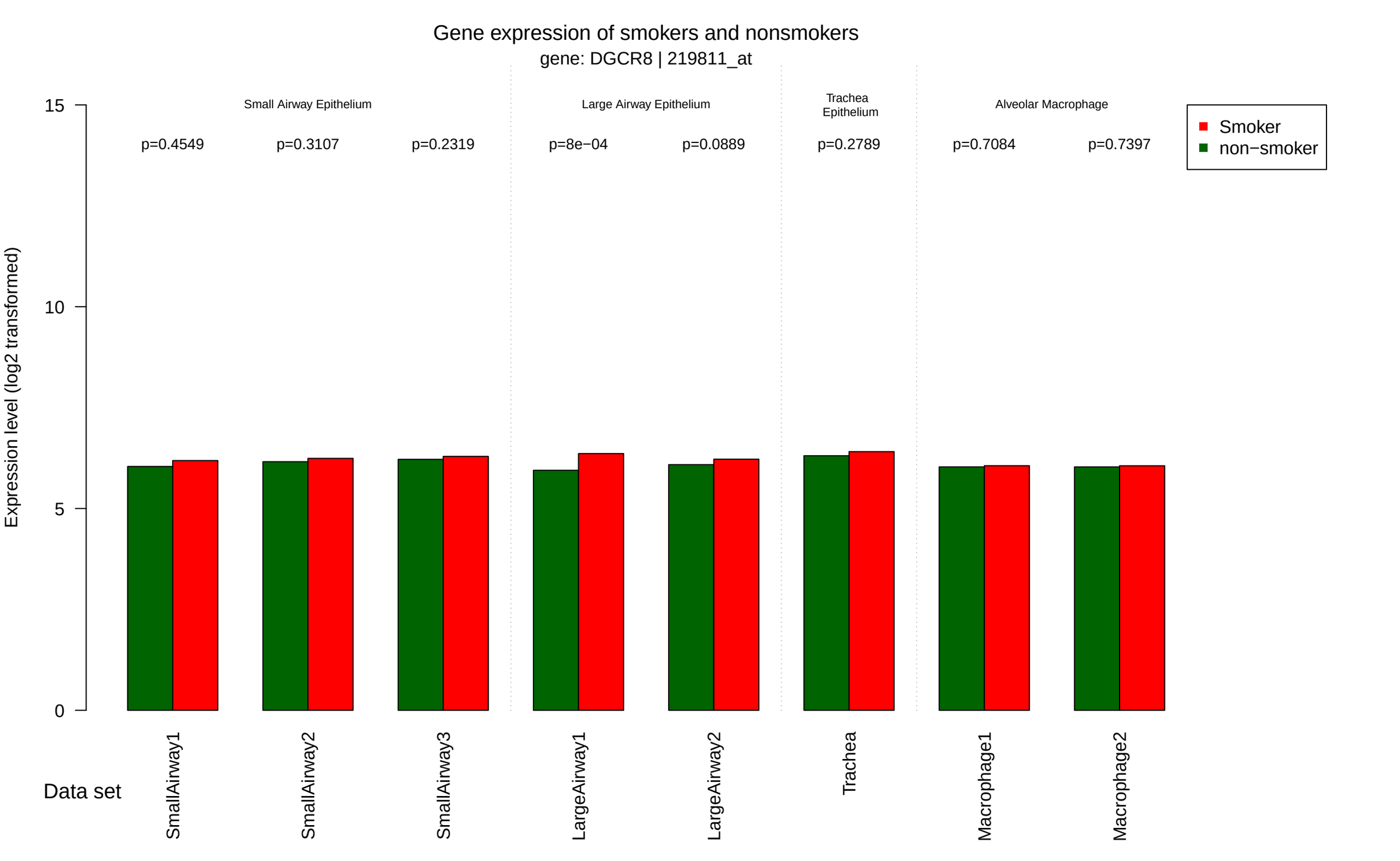
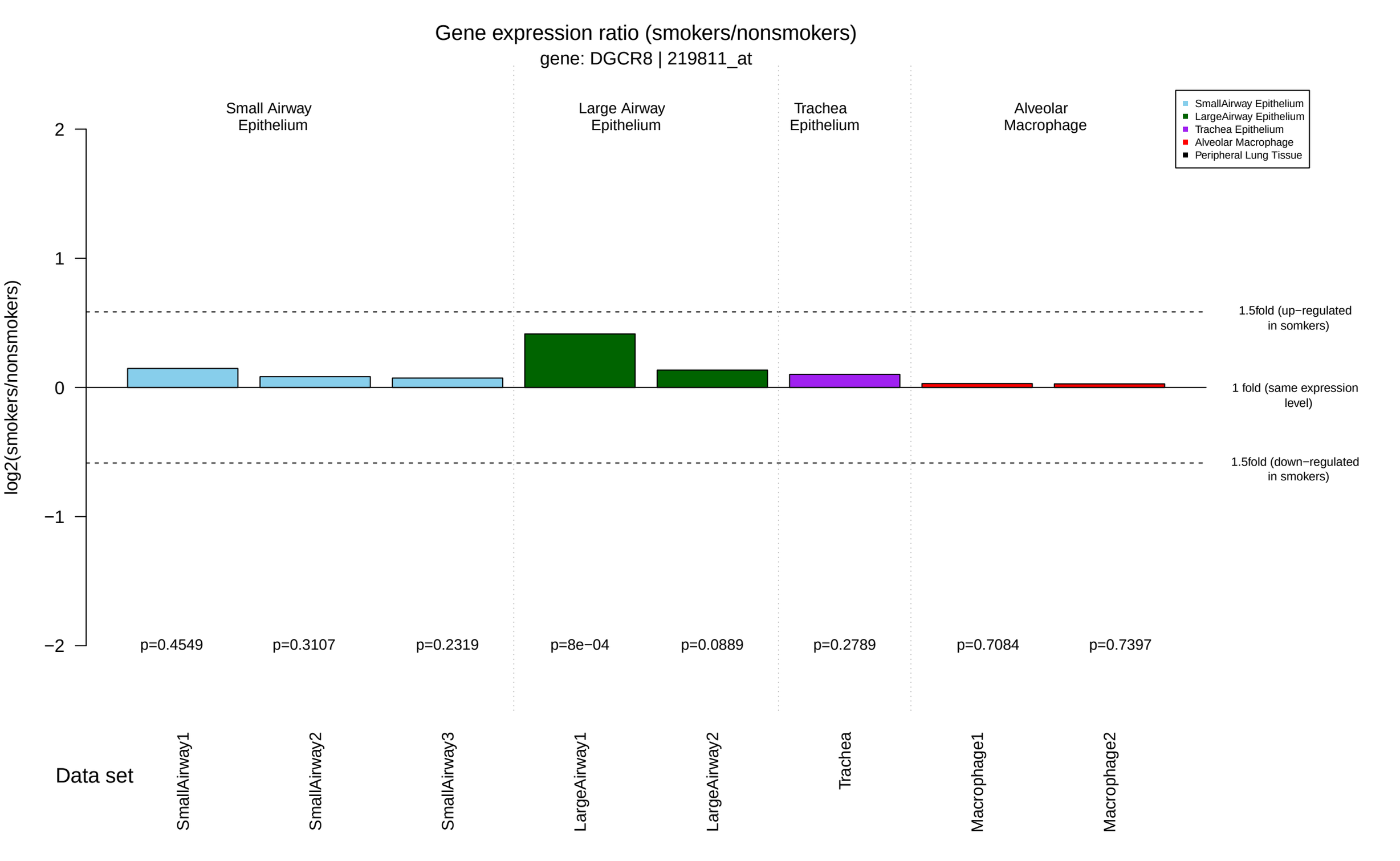
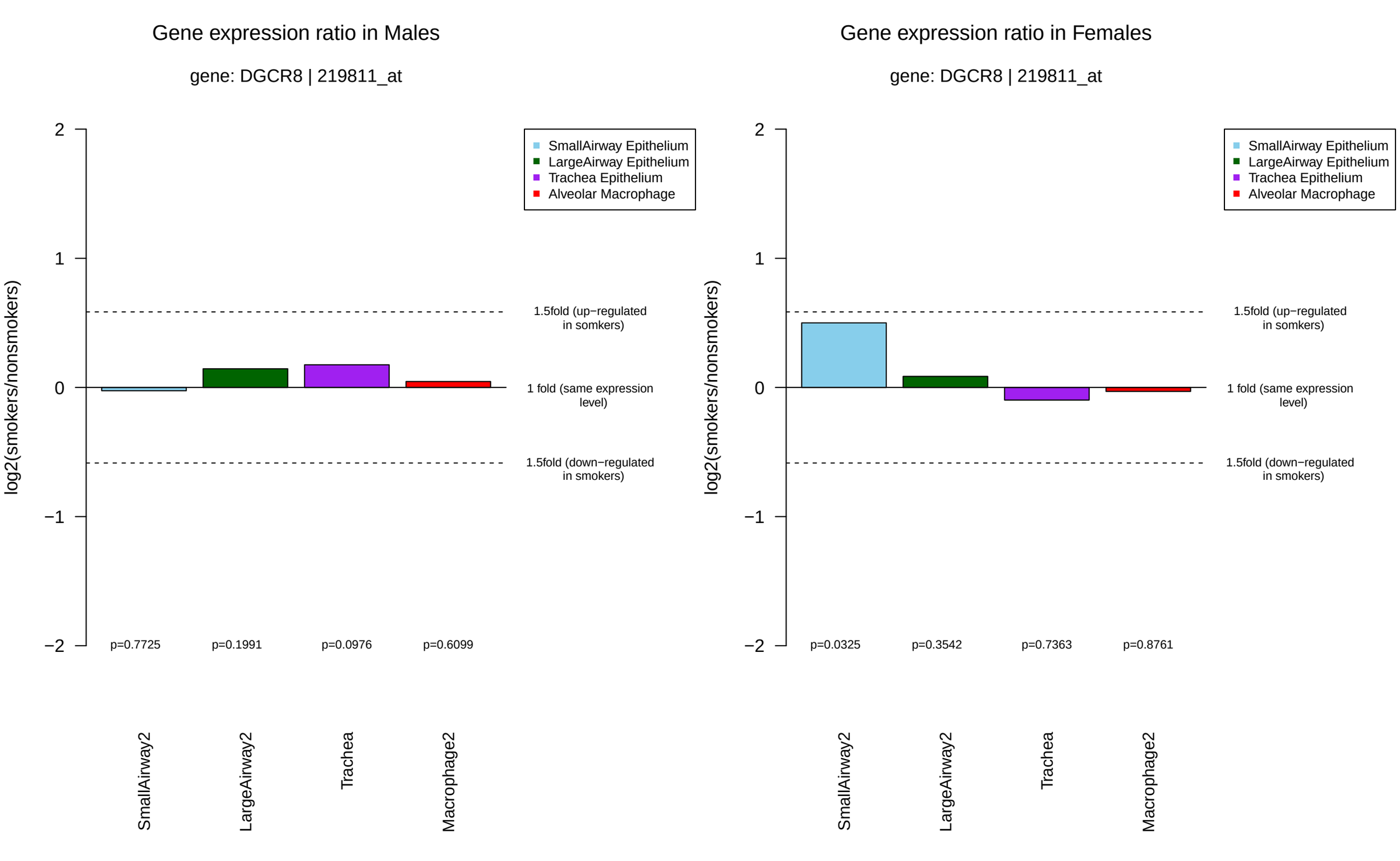
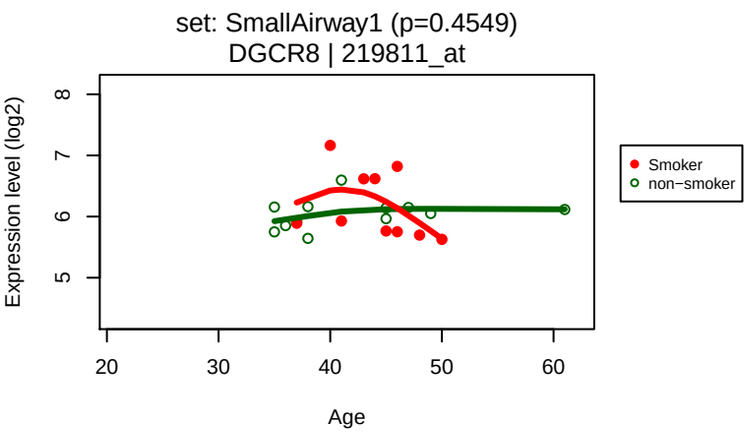
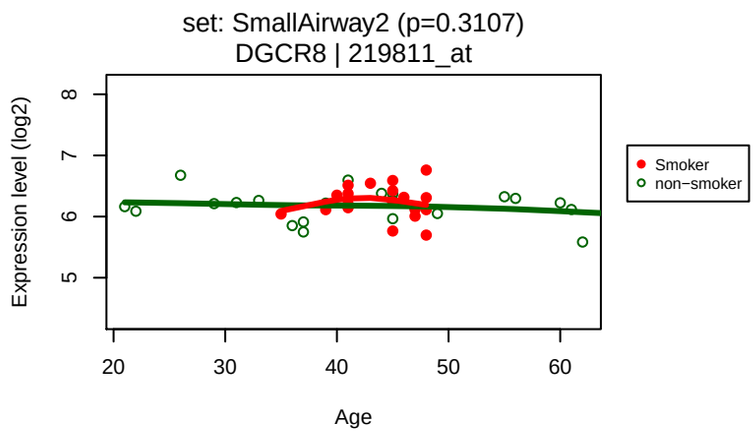
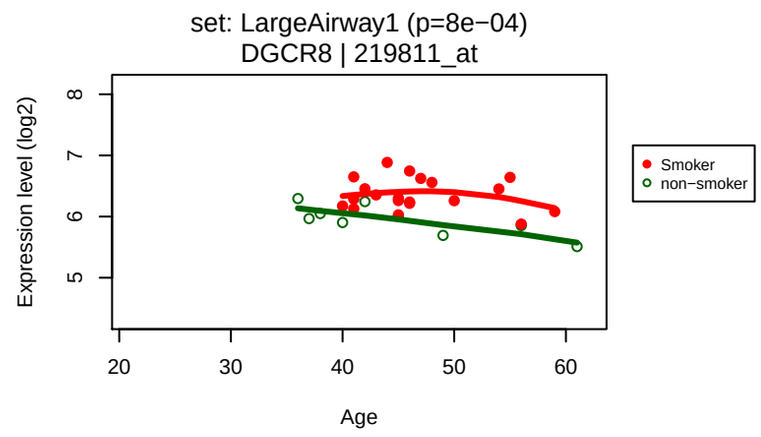

Gene expression of smokers and nonsmokers
gene: DGCR8 | 219811_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4549
p=0.3107
p=0.2319
p=8e−04
p=0.0889
p=0.2789
p=0.7084
p=0.7397
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

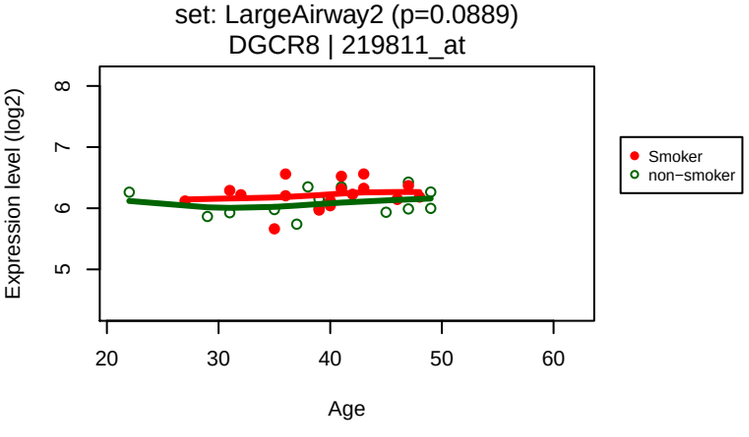

Gene expression ratio (smokers/nonsmokers)
gene: DGCR8 | 219811_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4549
p=0.3107
p=0.2319
p=8e−04
p=0.0889
p=0.2789
p=0.7084
p=0.7397
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

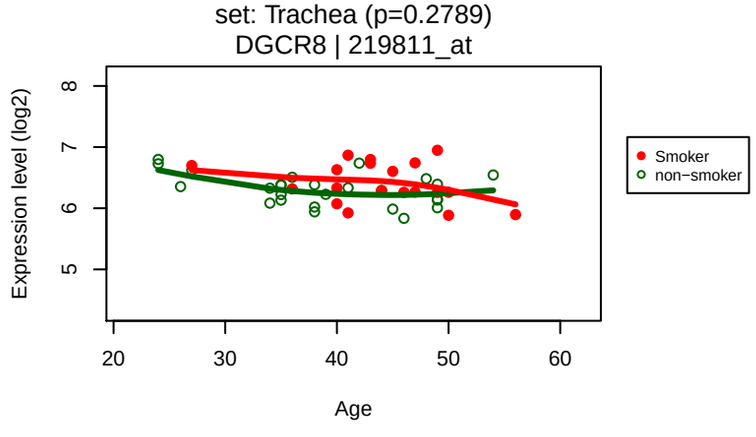

Gene expression ratio in Males
gene: DGCR8 | 219811_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7725
p=0.1991
p=0.0976
p=0.6099
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DGCR8 | 219811_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0325
p=0.3542
p=0.7363
p=0.8761
SmallAirway2
LargeAirway2
Trachea
Macrophage2

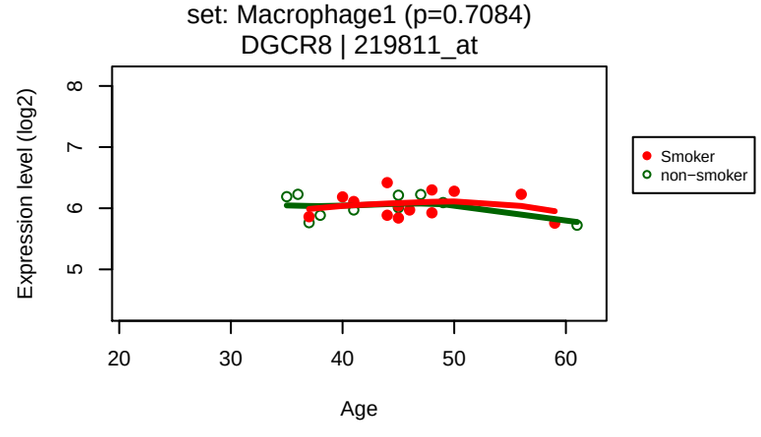

set: SmallAirway1 (p=0.4549)
DGCR8 | 219811_at
Expression level (log2)
Age
Smoker
non−smoker

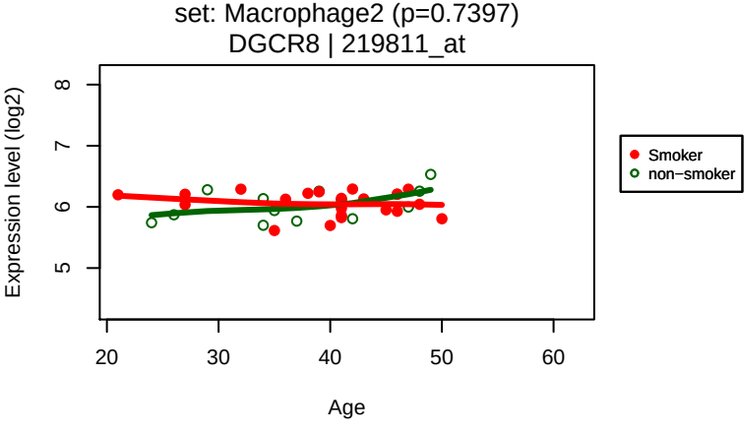

set: SmallAirway2 (p=0.3107)
DGCR8 | 219811_at
Expression level (log2)
Age
Smoker
non−smoker

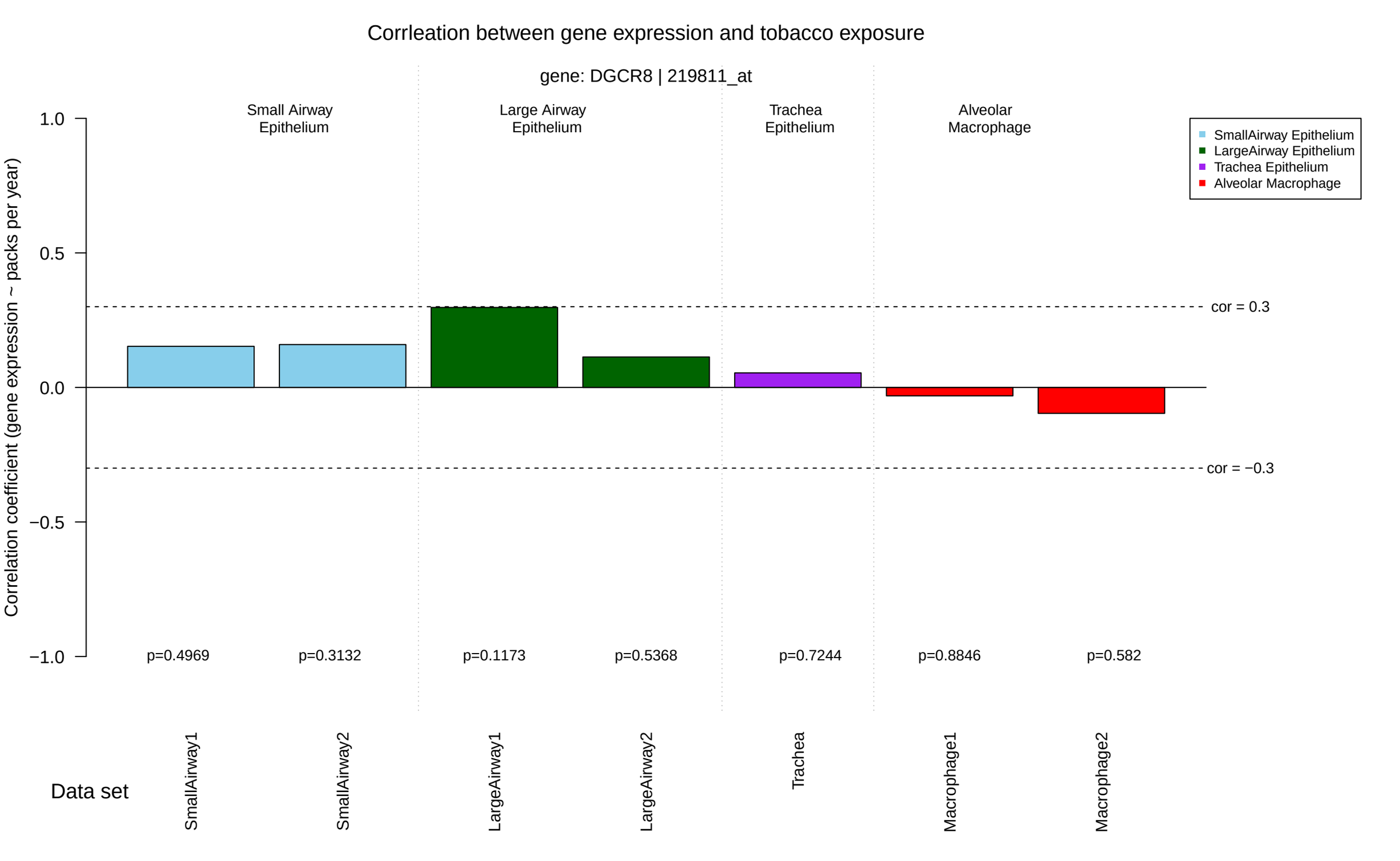

set: LargeAirway1 (p=8e−04)
DGCR8 | 219811_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0889)
DGCR8 | 219811_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2789)
DGCR8 | 219811_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7084)
DGCR8 | 219811_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7397)
DGCR8 | 219811_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DGCR8 | 219811_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4969
p=0.3132
p=0.1173
p=0.5368
p=0.7244
p=0.8846
p=0.582
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2